BRANCH/CLASS/BATCH: AIDS/A1/A
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Experiment No 8
Study of Path Loss in an Indoor Environment — RSSI vs. Obstacles

1. Aim

To study **path loss in an indoor wireless communication environment** by measuring **Received Signal Strength Indicator (RSSI)** at different distances and in the presence of different obstacles.

2. Objectives

1. To understand the concept of **wireless path loss**.
2. To understand the meaning and significance of **RSSI**.
3. To measure RSSI at different locations inside a building.
4. To study the effect of obstacles such as walls, doors, furniture, and people on signal strength.
5. To compare RSSI values for **line-of-sight (LOS)** and **non-line-of-sight (NLOS)** conditions.
6. To relate distance and obstacles to wireless signal attenuation.
7. To analyze the experimental data and identify the location with maximum and minimum signal strength.

3. Theory :

3.1 Wireless Signal Propagation

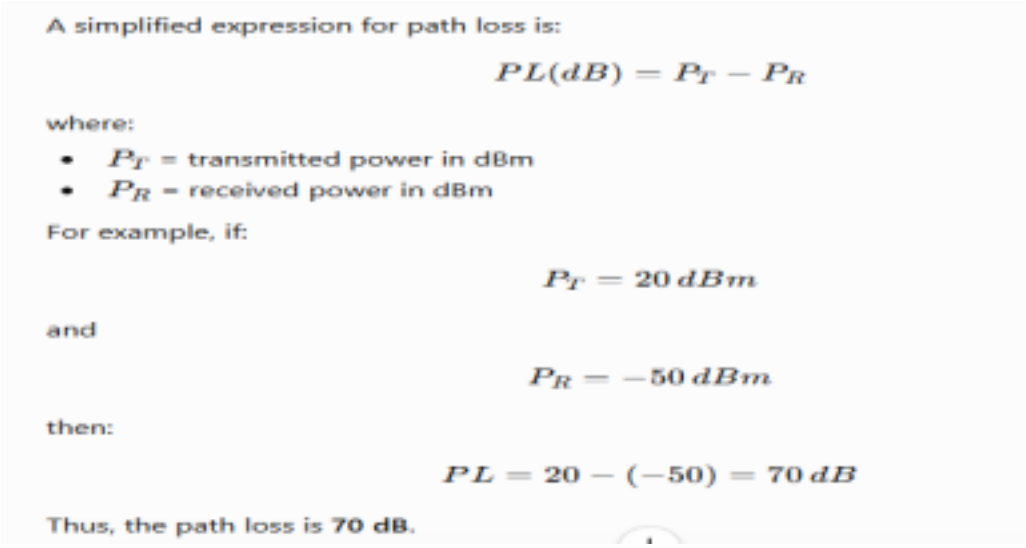
In a wireless communication system, information is transmitted from a **transmitter** to a **receiver** using electromagnetic waves.

For example:
Wi-Fi Access Point → Wireless Signal → Smartphone

As the signal travels through space, its power decreases. This reduction in received signal power is called **path loss**.
In an indoor environment, the signal does not always travel through a clear straight path. It may encounter walls, doors, furniture, human bodies, and other objects. Therefore, indoor wireless propagation is more complex than propagation in free space.

4. What is Path Loss?

Path loss is the reduction in power density or received signal power as a wireless signal travels from the transmitter to the receiver.
It is usually expressed in **decibels (dB)**.



A simplified expression for path loss is:

$$PL(dB) = P_T - P_R$$

where:

- P_T = transmitted power in dBm
- P_R = received power in dBm

For example, if:

$$P_T = 20\,dBm$$

and

$$P_R = -50\,dBm$$

then:

$$PL = 20 - (-50) = 70\,dB$$

Thus, the path loss is **70 dB**.

5. What is RSSI?
RSSI stands for Received Signal Strength Indicator.
RSSI is a measurement or indication of the **received wireless signal strength** at the receiver. For Wi-Fi measurements, RSSI is commonly represented in **dBm**.

Typical values may look like:
RSSI General Interpretation

Around -30 dBm Very strong
Around -50 dBm Strong
Around -60 dBm Good
Around -70 dBm Weak/moderate
Around -80 dBm Very weak
Around -90 dBm Extremely weak

These ranges are only general guidelines. The exact interpretation depends on the device, wireless technology, application, and environment.

Important point
RSSI is normally represented using negative dBm values.
Therefore:
−40 dBm is stronger than −70 dBm.
The value closer to zero represents the stronger received signal.

7. Relationship Between RSSI and Distance
As the distance between transmitter and receiver increases, the received signal generally decreases.

For example:
Distance Possible RSSI
1 m −40 dBm
3 m −48 dBm
5 m −55 dBm
10 m −65 dBm
15 m −72 dBm

These are **illustrative values only**. Students must record actual measured RSSI values.

Therefore:

Distance ↑ → Received Power ↓ → RSSI becomes more negative

6. Effect of Obstacles on RSSI
Indoor environments contain many obstacles.

Examples:
• Brick walls
• Concrete walls
• Glass
• Wooden doors
• Metal doors
• Furniture
• Human bodies
When a wireless signal encounters an obstacle, part of the signal may be:

• Absorbed
• Reflected
• Scattered
• Diffracted

As a result, the received signal strength can decrease.

For example:
Without wall:
Transmitter ——————————— Receiver
With wall:
Transmitter ——— | WALL | ——— Receiver
The receiver may observe a lower RSSI in the second case.

7. Line-of-Sight and Non-Line-of-Sight
Line-of-Sight (LOS)
When there is a relatively clear propagation path between transmitter and receiver, it is called

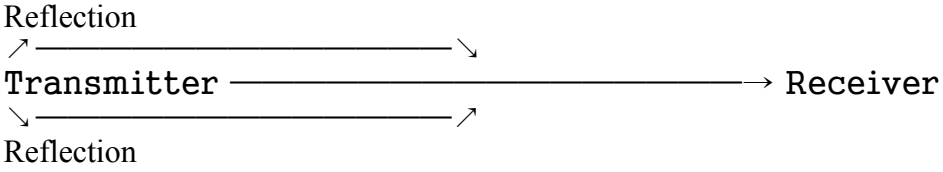
Line-of-Sight condition.
Access Point ——————————— Smartphone
Clear path

Non-Line-of-Sight (NLOS)
When an obstacle blocks the direct path between transmitter and receiver, it is called a **Non**

Line-of-Sight condition.
Access Point ———— | WALL | ———— Smartphone
Generally, NLOS conditions result in greater attenuation, although indoor multipath effects can make individual measurements fluctuate.

8. Indoor Propagation and Multipath
Indoor wireless signals can reach the receiver through multiple paths because of **reflection, scattering, and diffraction**.
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For example:



The receiver may therefore receive multiple versions of the same signal. These multiple signal components can combine constructively or destructively, causing **RSSI fluctuations** at different locations.
This phenomenon is known as **multipath propagation**.

9. Free-Space Path Loss
For an ideal free-space environment, path loss can be represented by:

For an ideal free-space environment, path loss can be represented by:

$$PL(dB) = 20\log_{10}\left(\frac{4\pi d}{\lambda}\right)$$

where:

- d = distance between transmitter and receiver
- λ = wavelength of the signal

In an actual indoor environment, the measured path loss can differ considerably from the free-space value because of walls, furniture, reflections, and other factors.

10. Log-Distance Path Loss Model
For indoor environments, a commonly used model is:

$$PL(d) = PL(d_0) + 10n\log_{10}\left(\frac{d}{d_0}\right) + X_\sigma$$

where:

- $PL(d)$ = path loss at distance d
- $PL(d_0)$ = path loss at reference distance d_0
- n = path-loss exponent
- X_σ = random variation caused by shadowing/environmental effects

The path-loss exponent **nn** describes how rapidly signal strength decreases with distance in a particular environment.

11.Observation Table

Distance	No Obstacle (LOS)	One Wall	Two Walls
2 m	-38 dBm	-45 dBm	-51 dBm
4 m	-46 dBm	-53 dBm	-60 dBm
8 m	-58 dBm	-66 dBm	-73 dBm
12 m	-65 dBm	-73 dBm	-81 dBm

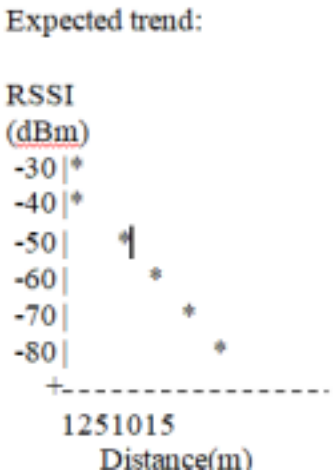
16 m	-72 dBm	-80 dBm	-88 dBm

From such observations, students can see that the RSSI generally becomes more negative when distance or obstruction increases.

12. Analysis
1. As the distance increases, RSSI generally decreases.
2. An RSSI value of **−50 dBm is stronger than −70 dBm**.
3. Obstacles generally reduce the received signal strength.
4. Concrete walls may cause greater attenuation than relatively less obstructive materials, but the exact result depends on thickness, construction, frequency, and environment. 5. RSSI may fluctuate even when the receiver is stationary because of multipath propagation and environmental changes**.**

Eg. Graph 1: RSSI vs Distance

• **X-axis:** Distance in metres
• **Y-axis:** RSSI in dBm



The graph generally shows a decreasing trend in received signal strength as distance increases.

Graph 2: RSSI vs Obstacles

• **X-axis:** Type of obstacle
• **Y-axis:** RSSI in dBm

This allows students to compare the effect of different obstacles.

13. Result

The path loss of a wireless signal in an indoor environment was studied by measuring **RSSI at different distances and under different obstacle conditions**.

It was observed that:

• RSSI generally decreases as distance increases.
• Obstacles cause additional signal attenuation.

• NLOS conditions generally produce weaker RSSI than clear-path conditions. • RSSI can fluctuate due to indoor multipath propagation.

Maximum measured RSSI = __-38____ dBm

Minimum measured RSSI = _-88_____ dBm

Maximum tested distance = ___16___ m

Output Readings

Parameter	**Observed Value**
Wi-Fi Frequency	2.4 GHz
Access Point	TP-Link Archer C6
Maximum RSSI	-38 dBm
Minimum RSSI	-88 dBm
Maximum Tested Distance	16 m
Strongest Location	2 m (LOS)
Weakest Location	16 m (Two Walls)

14. Applications

The study of indoor path loss and RSSI is useful in:

1. Wi-Fi network planning
2. Wireless LAN optimization
3. Indoor positioning systems
4. IoT networks
5. Smart buildings
6. Bluetooth and Wi-Fi communication
7. Wireless sensor networks
8. Network coverage analysis
9. Robot navigation
10. Location-based services

15. Conclusion

The experiment demonstrates that **indoor wireless signal strength is affected by both distance and physical obstacles**. RSSI provides a convenient way to observe the received signal strength at different locations.
The experiment also demonstrates that indoor environments are affected by **reflection, scattering, diffraction, and multipath propagation**, which can cause variations in RSSI.
This experiment is useful for understanding **Wi-Fi planning, wireless network optimization, IoT communication, indoor positioning, and network performance analysis**.